Section 4.2- Virtual Cell

Building World Model and Simulating Data for Molecular Medicine

2026 Spring

LLM Agents Foundation & Applications

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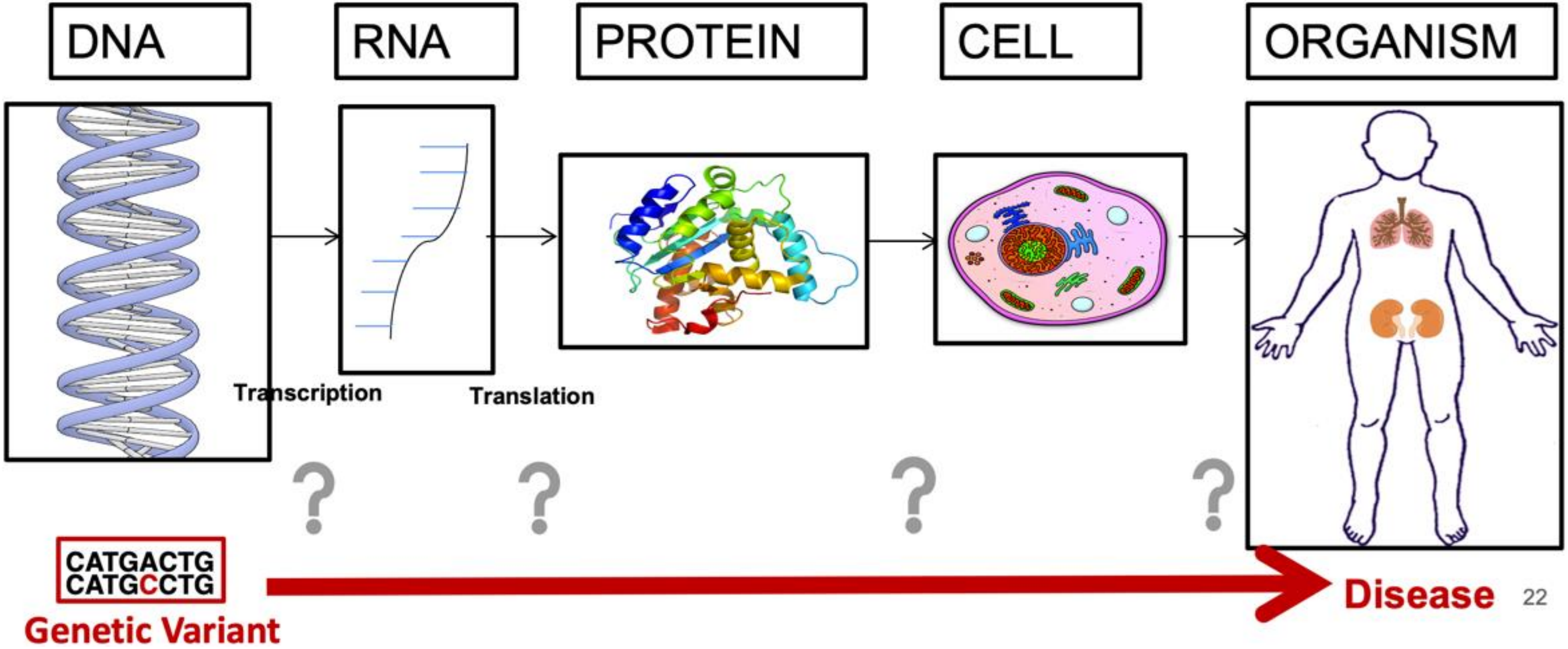
20260210

Biology in a Slide:



For Predictive Biology

THE GOAL

To build 'Virtual Cells'—AI models capable of predicting cellular responses to perturbations indistinguishably from

THE BOTTLENECK

Progress is stalled by a lack of standardized evaluations and high-quality, reproducible ground-truth data.

THE INITIATIVE

The Virtual Cell Challenge (VCC)—an annual, open-source competition launching in 2025 to evaluate model generalization.

THE ASSET

Powered by a purpose-built, high-depth H1 hESC dataset generated specifically for this challenge by the Arc Institute.

How to Build the AI Virtual Cell with Artificial Intelligence

How to Build the Virtual Cell with Artificial Intelligence: Priorities and Opportunities

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Why Model the Cell?

- The cell = fundamental unit of life
- Disease emerges from cellular dysfunction
- Current models:
  - Rule-based
  - Mechanistic but narrow
  - Hard to scale across modalities

Problem: Biology is multiscale, nonlinear, massively interactive

Limitations of Traditional Whole-Cell Models

- Differential equations
- Stochastic simulations
- Agent-based models

Work well for:

- Specific pathways
- Bacterial organisms

Struggle with:

- Multiscale integration
- Human cellular complexity
- Cross-modality reasoning

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Omics Revolution

- Single-cell RNA-seq
- Spatial transcriptomics
- Imaging
- Multi-modal atlases

AI Revolution

- Foundation models
- Transformers
- Diffusion models
- Generative modeling

Vision: The AI Virtual Cell (AIVC)

A learned, multi-scale, multi-modal neural simulator of cells and tissues

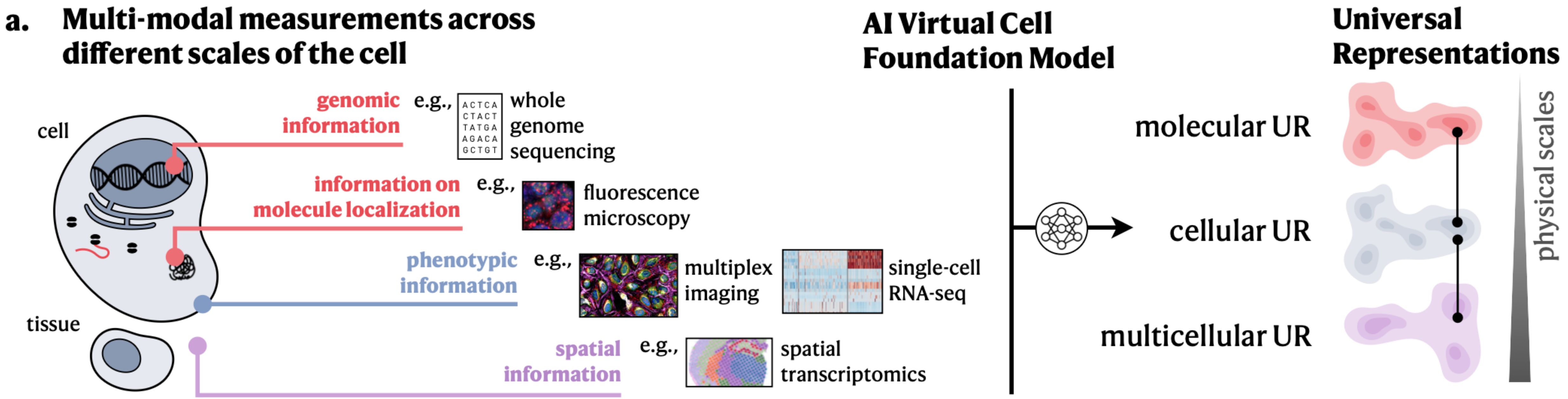
Core capabilities:

1. Universal biological representations
2. Predict behavior and dynamics
3. Enable in silico experimentation

Universal Representations (URs)

Must generalize to unseen states

- Embedding of biological state
- Integrates:
  - Species
  - Modalities
  - Scales
  - Conditions



Three Physical Scales

1. Molecular UR
2. Cellular UR
3. Multicellular UR

Each scale builds on the previous

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Molecular Scale

Data:

- DNA
- RNA
- Proteins
- Metabolites

Modeling:

- Sequence transformers
- Language modeling
- Atomic-level models

Goal: Learn molecular grammar

Three Physical Scales

1. Molecular UR
2. Cellular UR
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Each scale builds on the previous

Cellular Scale

Data:

- scRNA-seq
- scATAC-seq
- Proteomics
- Imaging

Modeling:

- Multi-modal transformers
- CNNs / Vision Transformers
- Cross-modal integration

Output: Unified cell-state embedding

Three Physical Scales

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Each scale builds on the previous

Multicellular Scale

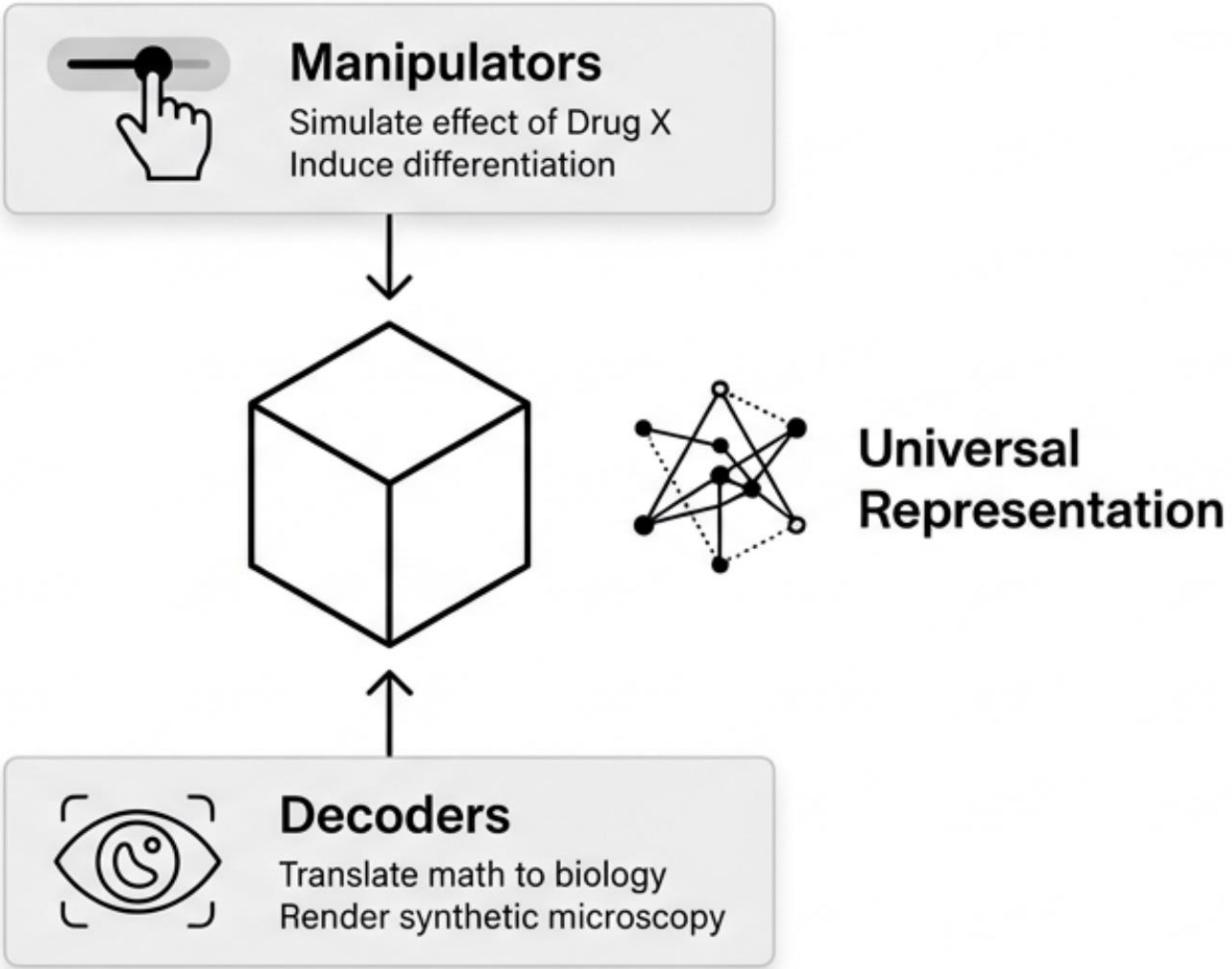
Data:

- Spatial transcriptomics
- Tissue imaging
- Spatial proteomics

Modeling:

- Graph neural networks
- Equivariant networks
- Spatial transformers

Goal: Model cell-cell interactions and niches



Manipulators
Simulate effect of Drug X
Induce differentiation
Universal Representation
Decoders
Translate math to biology
Render synthetic microscopy

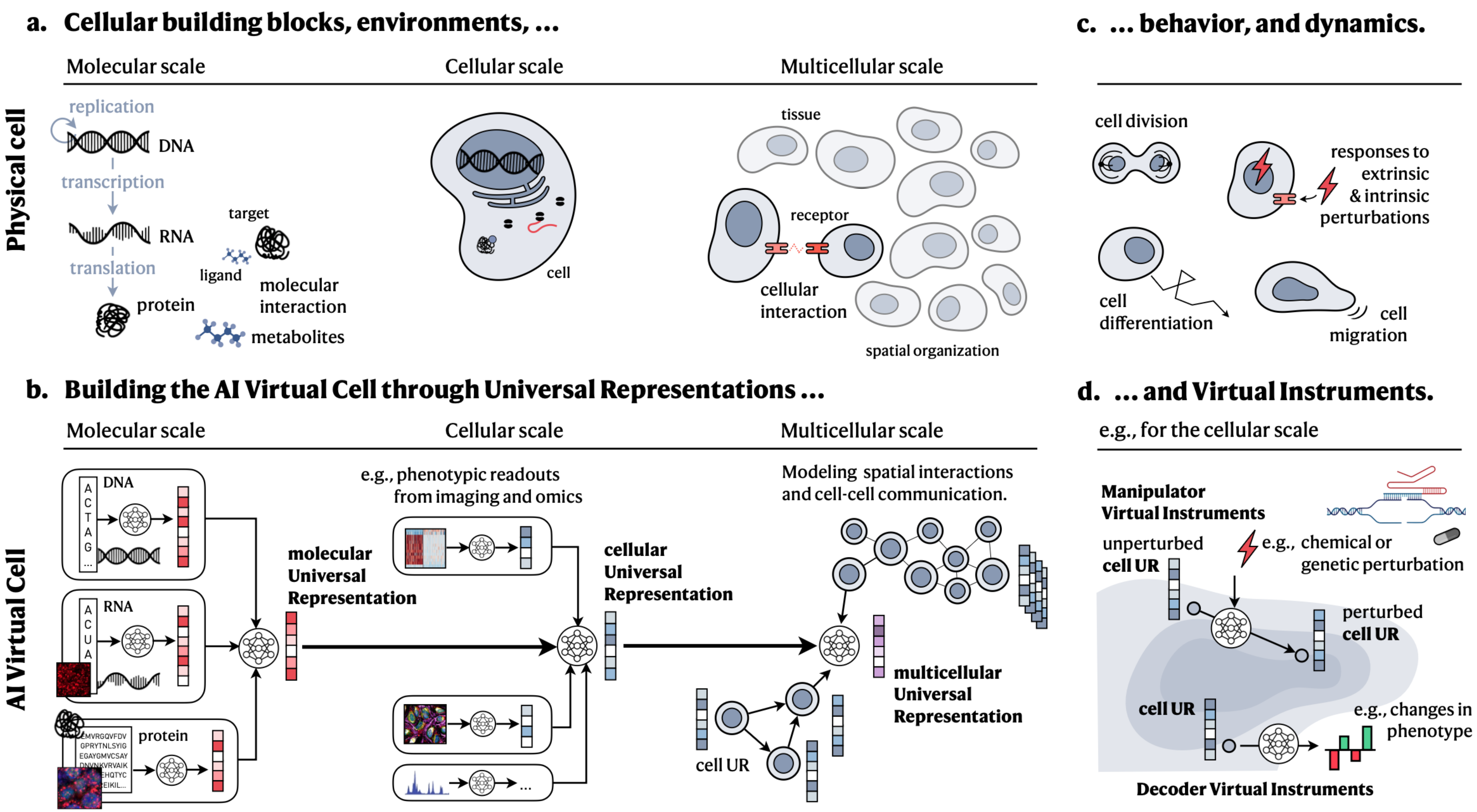


Figure 2: Overview of the AI Virtual Cell. a. Similar to biological cells, **b.** the AI Virtual Cell models cell biology across different physical scales, including molecular, cellular, and multicellular. Along the physical dimension, the first scale models the state and interactions of individual molecules, such as those of the central dogma, as well as additional molecules like metabolites. Molecules

Data Requirements

Must capture:

- Multi-scale biology
- Multi-modal signals
- Perturbations
- Temporal dynamics
- Human diversity

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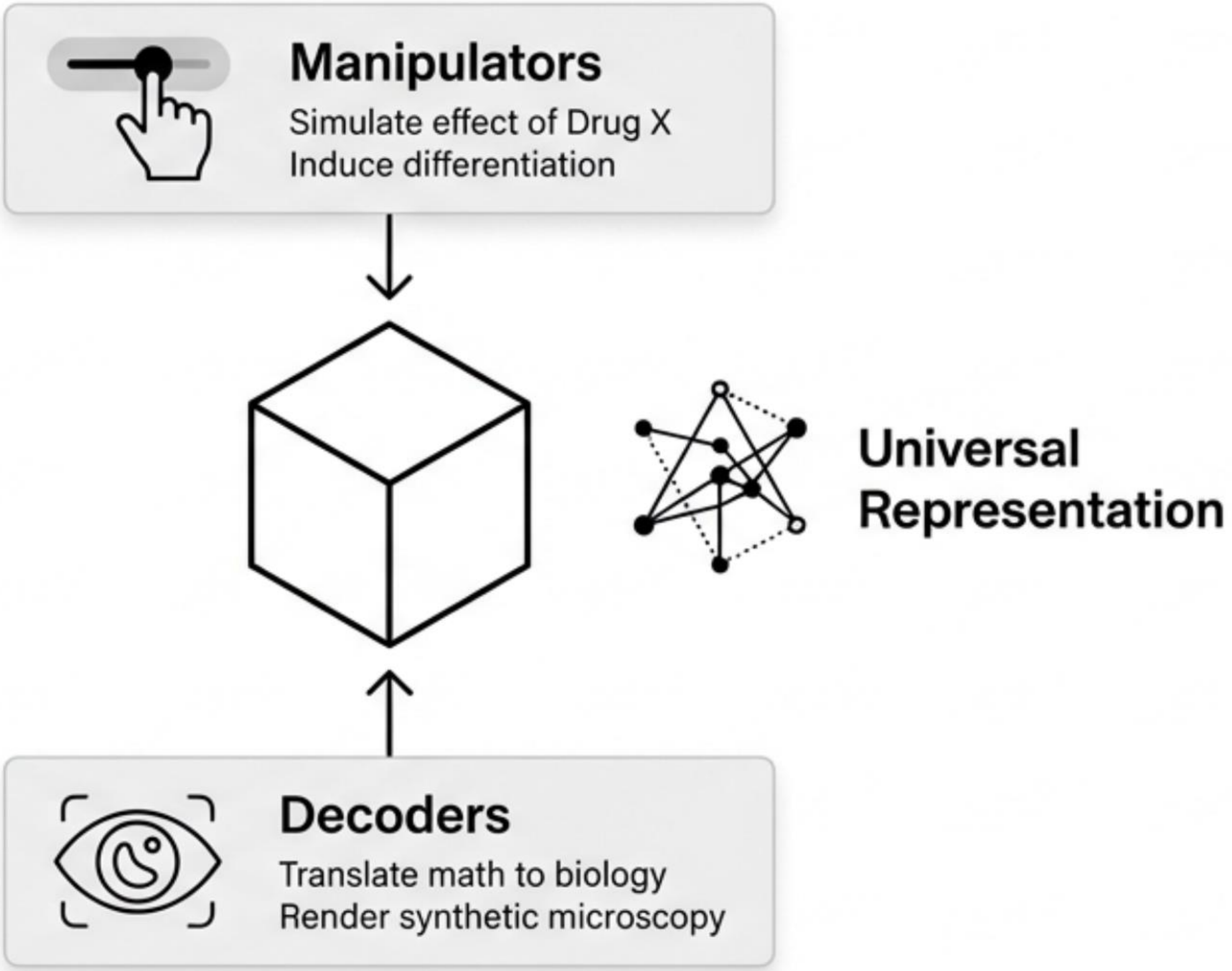
Data Challenges

- Species bias
- Population bias
- Technical noise vs biological variation
- Combinatorial explosion
- Unknown scaling laws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Virtual instruments are neural networks that manipulate or decode these representations.

Virtual Instruments (VIs)

Decoder VIs

- UR → phenotype / image / label

Manipulator VIs

- UR → modified UR
- Simulate perturbations

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Modeling Dynamics with Manipulators

Predict:

- Differentiation
- Migration
- Disease progression
- Perturbation response

Methods:

- Diffusion models
- Autoregressive transformers
- Flow matching

AI Architectures: Transformers

Used for:

- DNA
- RNA
- Protein sequences
- scRNA-seq

Strength:

- Self-attention models gene interactions

CNNs & Vision Transformers

Used for:

- Microscopy
- Histology
- Spatial phenotyping

Strength:

- Spatial pattern recognition

Diffusion Models

Used for:

- Dynamic modeling
- Continuous trajectories
- Perturbation prediction

Strength:

- High-dimensional generative modeling

Graph Neural Networks

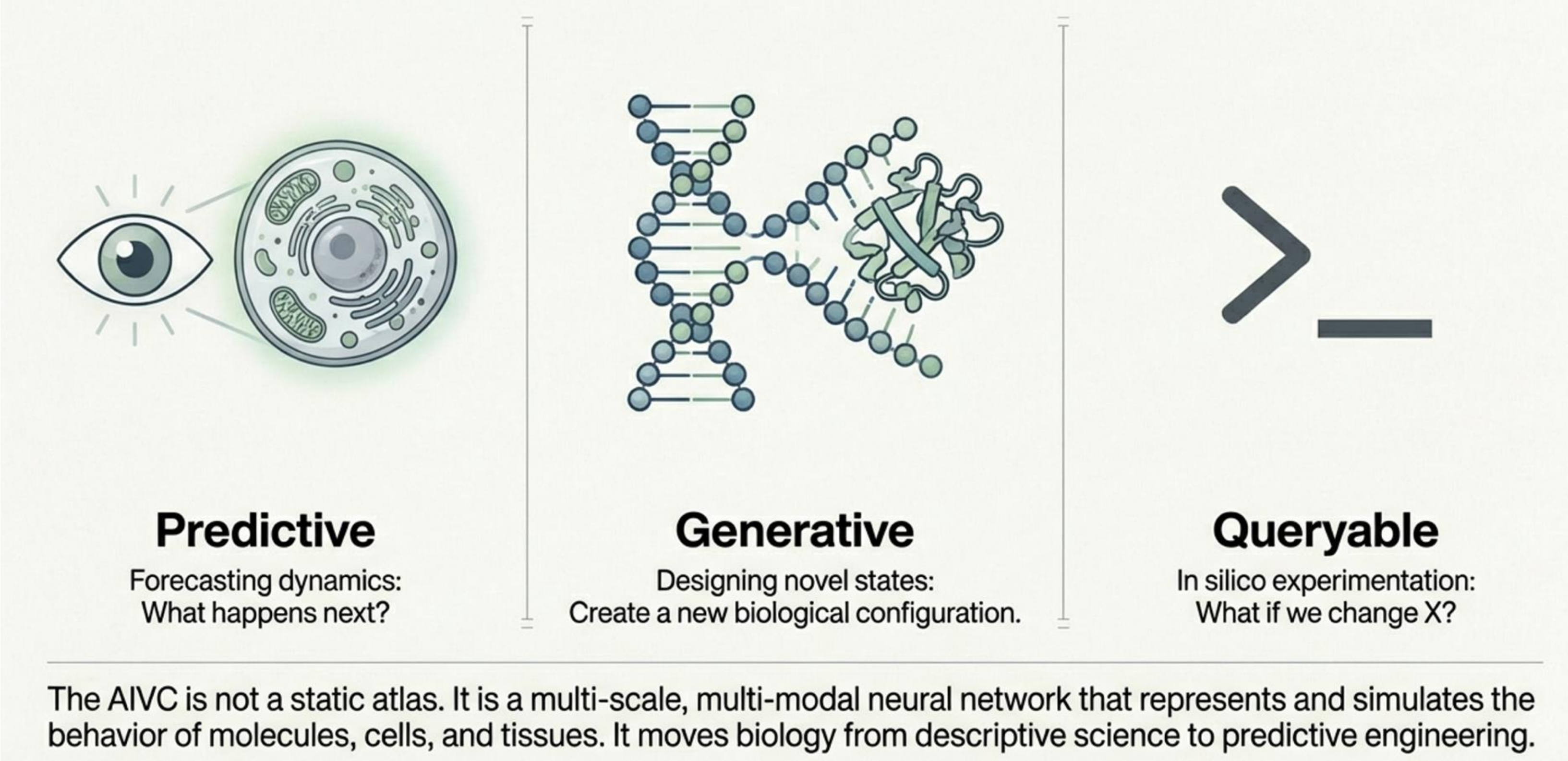
Used for:

- Tissue organization
- Protein structures
- Cell-cell communication

Strength:

- Structured relational modeling

Outlook



AI Virtual Cells could:

- Redefine biomedical research
- Enable programmable biology
- Transform drug discovery
- Create digital biological twins

A paradigm shift from modeling fragments → modeling life systems

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In Silico Experimentation

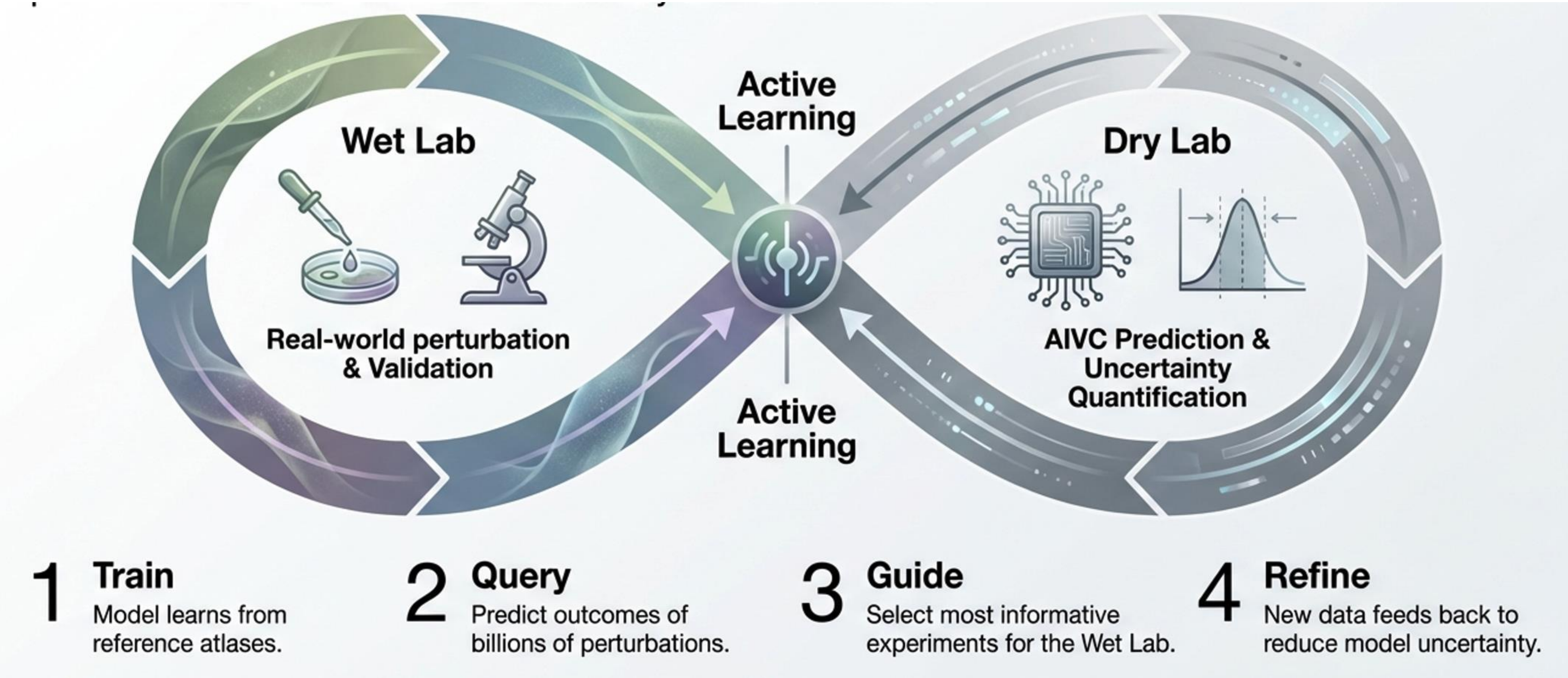
AIVC can:

- Screen combinatorial perturbations
- Simulate expensive assays
- Explore impossible experiments
- Guide real experiments

Creates a **lab-in-the-loop system**

Enable Lab in the loop

Lab-in-the-loop refers to an **active and iterative process** that creates a dynamic interaction between physical laboratory experimentation and AI-driven computational modeling. In this framework, the AI Virtual Cell (AIVC) acts as a "virtual laboratory" that facilitates a seamless interface with real-world experiments



Evaluating an AIVC

Metrics:

- Generalization to unseen contexts
- Cross-modal reconstruction
- OOD robustness
- Hypothesis generation ability

Beyond Accuracy: Biological Insight

AIVC should:

- Propose testable hypotheses
- Suggest causal mechanisms
- Link molecular changes → phenotype
- Reduce hypothesis search space

Application 1: Phenotypic Drug Discovery

- Simulate disease context
- Virtual phenotypic screens
- Prioritize interventions
- Accelerate cell-based therapies

Application 2: Spatial Oncology

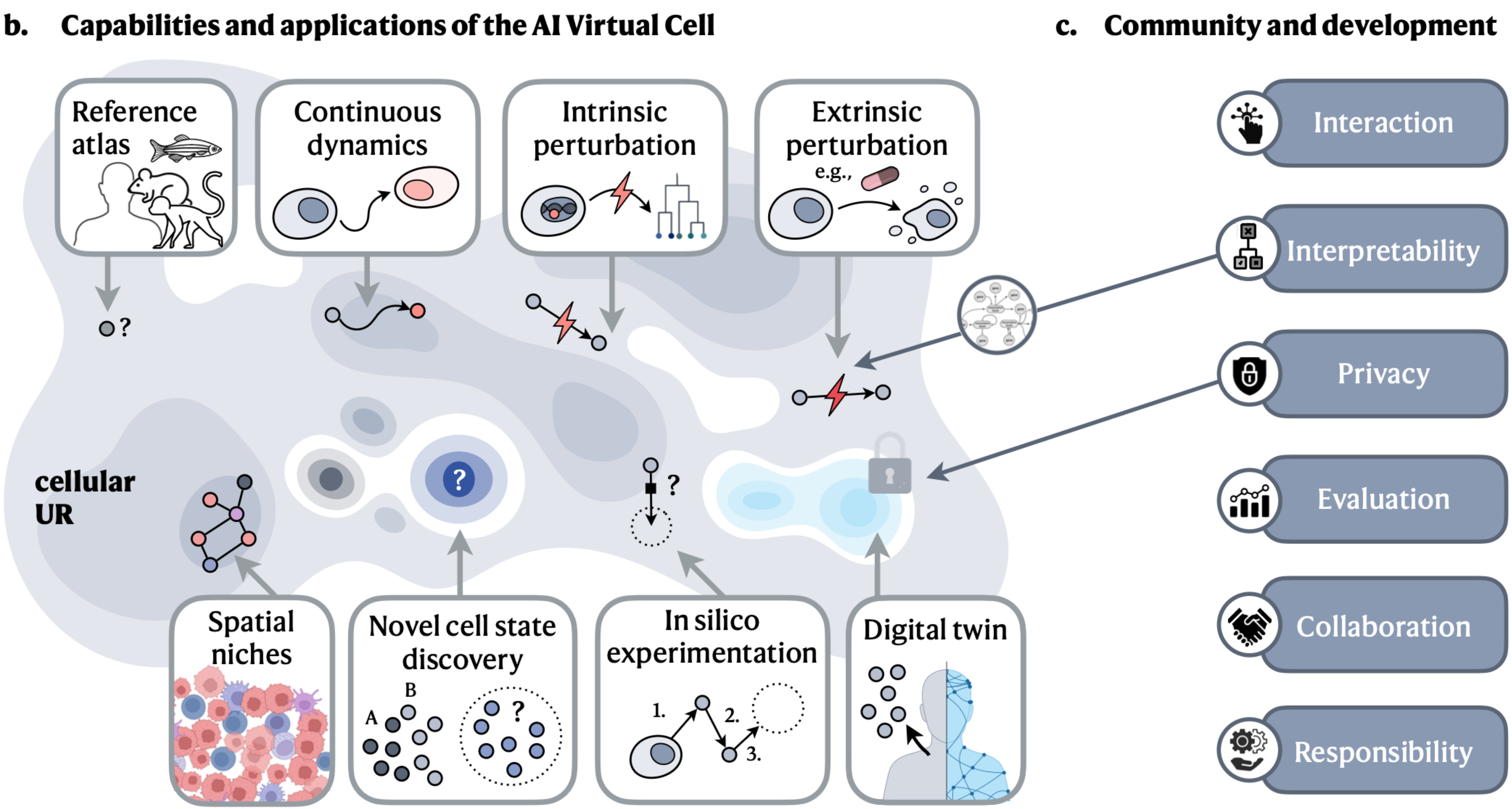
- Model tumor microenvironment
- Identify pan-cancer niches
- Discover resistance mechanisms
- Personalize treatment strategies

Application 3: Digital Twin Patients

- Patient-specific UR
- Integrate genomics + imaging + clinical data
- Predict disease progression
- Guide personalized therapy

Application 4: Hypothesis-Generating Engine

- Shift from validation → generation
- Active learning loop
- Toward self-driving biology labs



re 1: **Capabilities of the AI Virtual Cell. a.** The AI Virtual Cell provides a Universal Representation of a cell state that can be

Why Now?

- Massive cell atlases
- Biological foundation models
- Generative AI advances
- Cross-sector collaboration

Grand Challenges

1. Define core capabilities
2. Ensure self-consistency across scales
3. Balance interpretability vs utility
4. Prioritize data collection
5. Ensure ethical and equitable use
6. Build open collaborative infrastructure

Open questions:

- What is the minimal viable AIVC?
- How do we benchmark across scales?
- What inductive biases are essential?
- Should AIVCs be centralized or federated?

NEXT Paper

Virtual Cell Challenge

Toward a Turing Test for the Virtual Cell

Roohani, Hua, Tung et al. · Arc Institute · Cell, June 2025

What Is a "Virtual Cell"?

> A **virtual cell** is a computational model that learns the relationship between cell state and function, and predicts the consequences of perturbations—such as gene knockdowns or drug applications—across cell types and contexts.

Why does this matter?

- Understanding how cells respond to perturbations is central to basic biology and drug discovery
- Experimental testing of all perturbations × all cell types is **impossibly expensive**
- A model that can generalize across contexts could revolutionize biology and medicine

Why Do We Need a Benchmark for Virtual Cells?

Why Now? Conditions Have Changed

- Massively improved single-cell measurement technologies (scRNA-seq, CRISPRi screens)
- Richer perturbational datasets — genome-wide Perturb-seq at scale
- Advances in machine learning and foundation models (scGPT, GEARS)

But no standardized benchmarks exist to evaluate whether models capture generalizable biological structure vs. dataset-specific patterns.

Inspiration: CASP for Protein Folding

- CASP (1994–) created a recurring blind competition for protein structure prediction
- It catalyzed the field and ultimately enabled AlphaFold2 — which solved protein folding
- ImageNet catalyzed computer vision in the same way
- **The Virtual Cell Challenge aims to do the same for predictive cell biology**

Past Competitions and Remaining Gaps

Competition	What It Tested	Gap
Broad Cancer Immunotherapy Challenge	Broad phenotypic shifts in T cells	No gene expression resolution
NeurIPS-Kaggle 2023	Gene expression response to **small molecules**	Limited to chemicals; no genetic perturbations
Virtual Cell Challenge (2025)	Gene expression response to **genetic perturbations**	← *This work*

Why genetic perturbations?

- Precisely defined targets → ideal for probing causal gene-function relationships
- Transcriptional changes can be more subtle → harder to predict → better test

Task: Context Generalization

The key challenge: generalization across cell types

Task formulation:

- Participants have seen perturbation effects in **multiple other cell contexts**
- A **subset** of perturbation effects in H1 hESCs is provided (few-shot adaptation)
- Goal: predict the effects of **held-out perturbations** in H1 hESCs

Why few-shot, not zero-shot?

> Most published datasets span only a handful of cell lines — true zero-shot generalization to new cell states is likely premature at this stage.

Task Design: core components & Three Metrics

Task: Predict Gene Expression Response to CRISPRi Perturbations in H1 hESCs

Core components of the challenge:

- Annual, open competition hosted by Arc Institute
- Purpose-built high-quality benchmark dataset (H1 hESC line)
- Public leaderboard for real-time progress tracking
- Three complementary evaluation metrics
- Training data from the Arc Virtual Cell Atlas + public datasets

Website: virtualcellchallenge.org

The Three Evaluation Metrics

Combined score weighted across all three metrics. Minimum thresholds enforced on all metrics — prevents gaming by models that excel at one dimension at the expense of others.

Metric 1
Differential Expression Score
How accurately does the model predict DE genes?

Metric 2
Perturbation Discrimination Score
Can the model distinguish between perturbations?

Metric 3
Mean Absolute Error (MAE)
Global accuracy across all genes

Benchmark Dataset: High-Quality CRISPRi Screen

Dataset Generation Pipeline

- Start: Broad low-depth screen of 2,500 CRISPRi perturbations in H1 hESCs
- Select 300 perturbations covering broad range of effect sizes (strong → subtle → negligible) and diversity of phenotypic responses (clustering-based selection)
- High-depth targeted sequencing using 10x Genomics Flex chemistry (fixed-cell, reduced batch effects)

Competition Dataset Split

Training set

150 perturbations · ~150,000 cells · Released at launch

Validation set

50 perturbations · ~50,000 cells · Live leaderboard

Test set

100 perturbations · ~100,000 cells · Released 1 week before deadline · Final rankings

Additional Training Resources: Arc Virtual Cell Atlas — scBaseCount + Tahoe-100M (>100M cells, largest public perturbational scRNA-seq dataset). Combined: >350 million cells and counting.

The Virtual Cell Challenge benchmarks the hardest, most relevant real-world task:

The Benchmark in Context

What good performance would demonstrate:

- A model can generalize gene expression predictions to a new, unseen cell type (H1 hESC)
- Using only a **few-shot subset** of H1 hESC perturbations for adaptation
- Achieving balanced scores on all three complementary metrics
- Reproducing biologically meaningful differential expression signatures

Key Quality Metrics Achieved

Parameter	Value
Total cells profiled	~300,000
Perturbations profiled	300 genes (CRISPRi)
Median cells / perturbation	~1,000 cells
Mean reads per cell	~93,000
Median gene UMIs per cell	~50,000

Goals of the Virtual Cell Challenge

The challenge is designed to achieve three outcomes:

1. Surface best practices and data considerations

- Training set quality, sequencing depth, platform choice
- Guidelines for reproducible and principled data generation

2. Identify methodological bottlenecks

- Clarify which model and data choices drive generalization
- Converge on reproducible standards for perturbation modeling

3. Highlight the importance of data quality

- Cell coverage, sequencing depth, and technology choice
- By underscoring these issues, drive higher-quality community data generation

Future Directions

Expanding the challenge over time:

Near-term expansions:

- Combinatorial perturbations (gene × gene, gene × drug)
- Cross-cell-type generalization at scale
- Multi-modal predictions (proteomics, epigenomics)

Longer-term vision:

- Temporal and spatial dimensions
- Multicellular systems
- **Inverse design:** identify perturbations that achieve a desired cellular effect (therapeutic relevance)

> 🔄Metrics and datasets will evolve each year based on insights from prior challenge results — this is a **living benchmark**, not a static test.

The Broader Vision: A Turing Test for the Virtual Cell

> "Virtual cells are poised to become foundational tools for biology, and to ensure that they reach their potential, we need clear and rigorous evaluations."

Analogies:

- **CASP** → catalyzed protein structure prediction → AlphaFold
- **ImageNet** → catalyzed computer vision
- **Virtual Cell Challenge** → could catalyze predictive cell biology

The ultimate goal:

> A model indistinguishable from experiment — a true Turing test for the virtual cell

Summary

Component	Key Detail
Task	Predict gene expression response to genetic perturbations in H1 hESCs (context generalization)
Format	Few-shot: subset of H1 hESC perturbations provided; 100 held-out test perturbations
Dataset	300 CRISPRi perturbations × ~1,000 cells each; 10x Genomics Flex; high depth
Metrics	DE score + perturbation discrimination score + MAE (combined + minimum thresholds)
Training data	Arc Virtual Cell Atlas (>350M cells) + public perturbation datasets
Goal	Reproducible, fair benchmarking to surface generalizable models of cell behavior

Participate: virtualcellchallenge.org | **Data:** arcinstitute.org/tools/virtualcellatlas

Appendix: Key Definitions

CRISPRi (CRISPR interference): A tool to repress (silence) gene expression using a catalytically dead Cas9 fused to a transcriptional repressor, guided to a specific genomic locus by a guide RNA.

scFG (single-cell functional genomics): Coupling genetic perturbations with single-cell transcriptomics to measure how gene expression changes in response to each perturbation at single-cell resolution.

Perturb-seq: A widely used scFG approach combining pooled CRISPR perturbations with scRNA-seq profiling.

UMI (Unique Molecular Identifier): Molecular barcodes used in scRNA-seq to count individual RNA molecules, reducing PCR amplification bias.

Flex chemistry: 10x Genomics' fixed-cell scRNA-seq protocol that allows multiplexed sample pooling and reduces batch effects through cell fixation before library preparation.